



Dogs domesticated in ice age

Excess protein found in lean meats during harsh winters were used to domesticate dogs in the Ice Age. <http://digit.in/mar21-05>



Communication with dogs

An example of a scientific "spat", a study that is a response to a study that is a response. <http://digit.in/mar21-06>



How did the big bad wolf become man's best friend?

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ogs were the first animals to be domesticated, and the only animals to be domesticated by

hunter gatherers some time during the last ice age, between 30,000 and 12,000 years ago, before the advent of agriculture. The question of when and where the wolf was domesticated is a surprisingly intense field of research, with papers coming in from a wide variety of disciplines including archeology, neurochemistry and genetics. Many of these papers challenge and critique previous findings, reinterpret results, pry apart questionable methodologies, and may even be response study

to a response study. Scientists are fighting over this apparently simple question. During the last ice age, men hunted in bands, and wolves hunted in packs, both competed to take down large game such as horses, bison, caribou and if they were really skillful, a mammoth. Both men and wolves were capable of killing each other. How, why and where these two competing predators managed to cooperate is an enduring puzzle.

THE CHANGES IN THE BRAIN

In 1952, the Russian geneticist Dmitry Belyayev was fascinated by the wide variety of domestic dog breeds and wondered how they could have all evolved from wolves. He believed the answer to be in Mendelian inheritance, and that it was selective breeding that gave rise to all the domestic varieties. Belyayev suspected that when the domestication of dogs occurred, the primary guiding factor for selection was tameness, which means that humans domesticated the wolves by

selectively breeding the tamest and friendliest dogs. At that time research into genetics was frowned upon in Russia, so Belyayev designed an ingenious plan of using silver foxes for the study, while pretending to feed the fur industry and increase Russian production. Initially the experiment began with breeding two populations, one without any external pressure on selection, and one were the only selection criteria was tameness. In only a few generations, the foxes became increasingly similar to domestic dogs, with the pups approaching humans near cages, whining for attention, showing their bellies and wagging their tails. These changes were accompanied by physical changes as well, with the ears drooping, tails shortening, and changes in the shape of the skull. The study continues to this day, with the researchers studying the effects of selection for tameness on otters and minks as well.

The changes were also physiological, as Belyayev suspected with the first changes detected being lowered